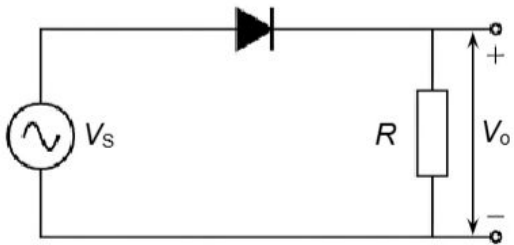
18 An ideal semiconductor diode is connected in series with a load resistor $R = 50\,\Omega$ and an alternating voltage source $V_S = 12.0\sin(100\pi t)$. The output voltage V_o is measured across the load resistor R.



Ans: B

The diode only conducts during the positive half-cycle, so current flows through R for only half the cycle.

Peak voltage across R:

$$V_0 = 12.0\ \text{V}$$

Average power over a full cycle: Since the diode conducts only for half the cycle, the average power is half that of a normal AC circuit:

$$\langle P \rangle = \frac{1}{2}\left(\frac{V_0^2}{2R}\right) = \frac{V_0^2}{4R} = \frac{12.0^2}{4 \times 50} = \frac{144}{200} = 0.72\ \text{W}$$

What is the average power dissipated in the load resistor R over one complete cycle?

A 0 W **B** 0.72 W **C** 1.44 W **D** 2.88 W